

Lab3: Insight into JavaScript and Bootstrap

Objective

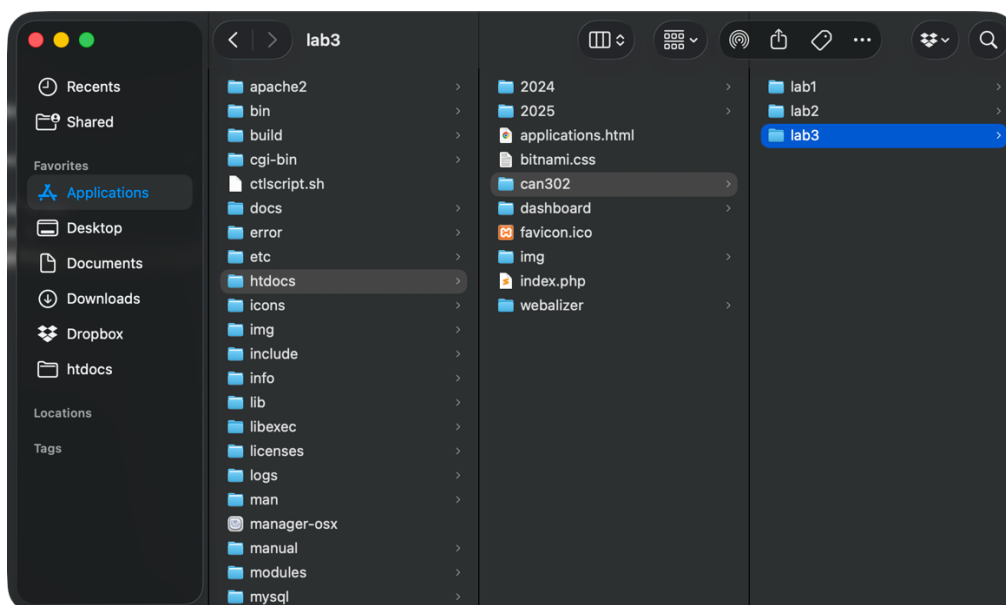
- Demonstrate the basic working knowledge of JavaScript
- Use JavaScript to update HTML content and handle HTML events
- Demonstrate the basic working knowledge of Bootstrap
- Use Bootstrap to create responsive webpages

Tips

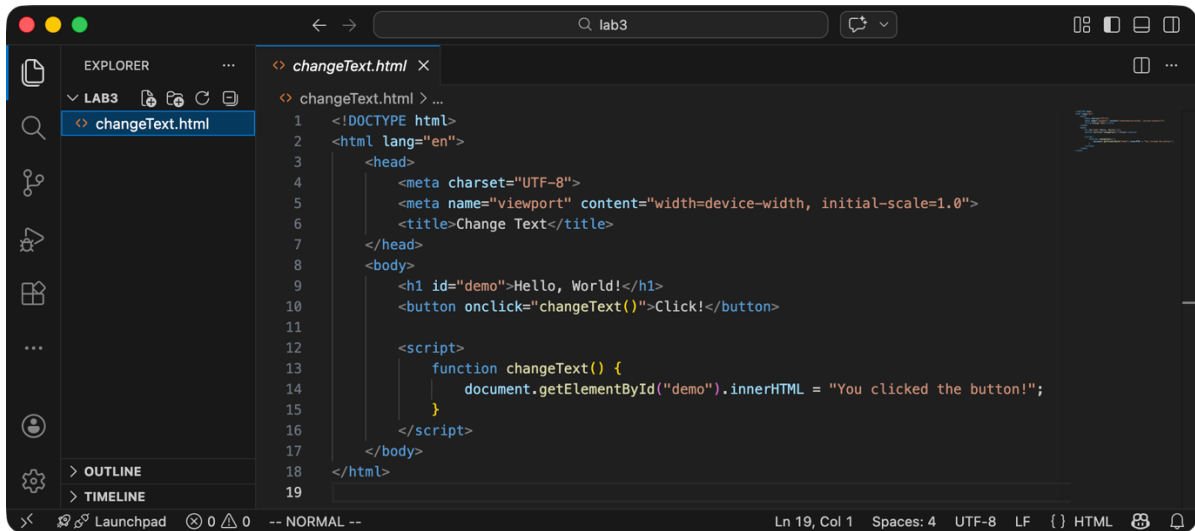
- These labs are expected to take more than the allocated 2 hours. You should try the tasks before the lab. Practice makes code perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

A. JavaScript Basics

1. Create a new subfolder "lab3" under folder "can 302" (The following screenshot is in macOS, but same operations can be applied to Windows):



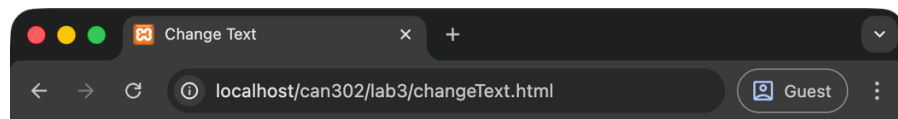
2. Use VS Code to open "lab3" and create a new file "changeText.html" as follows:



The screenshot shows the Visual Studio Code editor with a file named 'changeText.html' open. The code is as follows:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Change Text</title>
7   </head>
8   <body>
9     <h1 id="demo">Hello, World!</h1>
10    <button onclick="changeText()">Click!</button>
11
12    <script>
13      function changeText() {
14        document.getElementById("demo").innerHTML = "You clicked the button!";
15      }
16    </script>
17  </body>
18 </html>
19
```

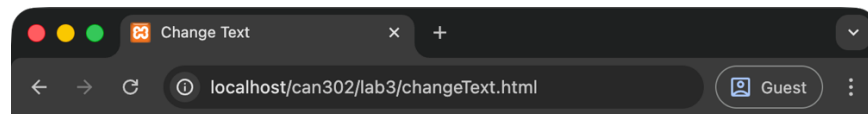
3. Access the webpage in your browser via URL: <http://localhost/can302/lab3/changeText.html>



Hello, World!

Click!

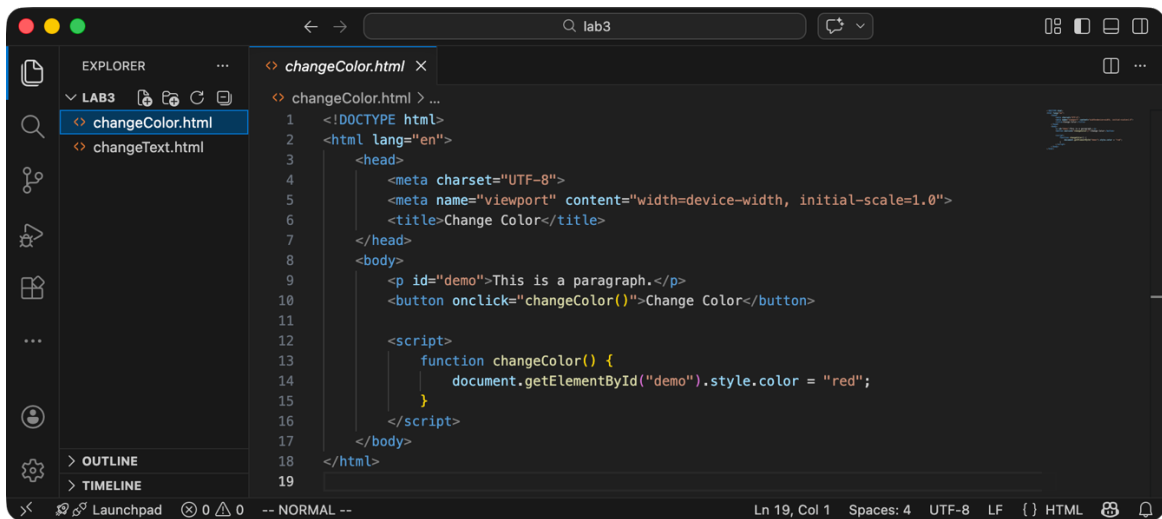
4. Click the button of "Click!", it will change to:



You clicked the button!

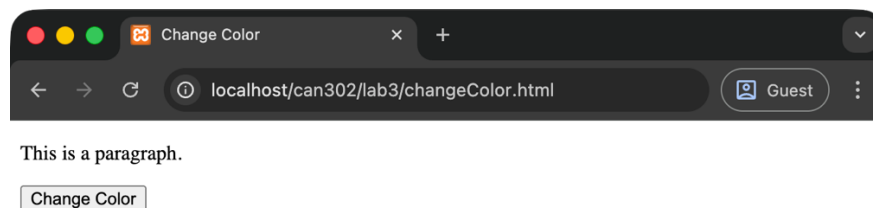
Click!

5. Create a new file "changeColor.html" under "lab3" as follows:

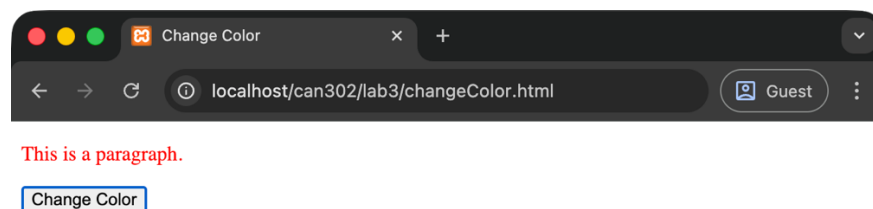


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Change Color</title>
7   </head>
8   <body>
9     <p id="demo">This is a paragraph.</p>
10    <button onclick="changeColor()">Change Color</button>
11
12    <script>
13      function changeColor() {
14        document.getElementById("demo").style.color = "red";
15      }
16    </script>
17  </body>
18 </html>
19
```

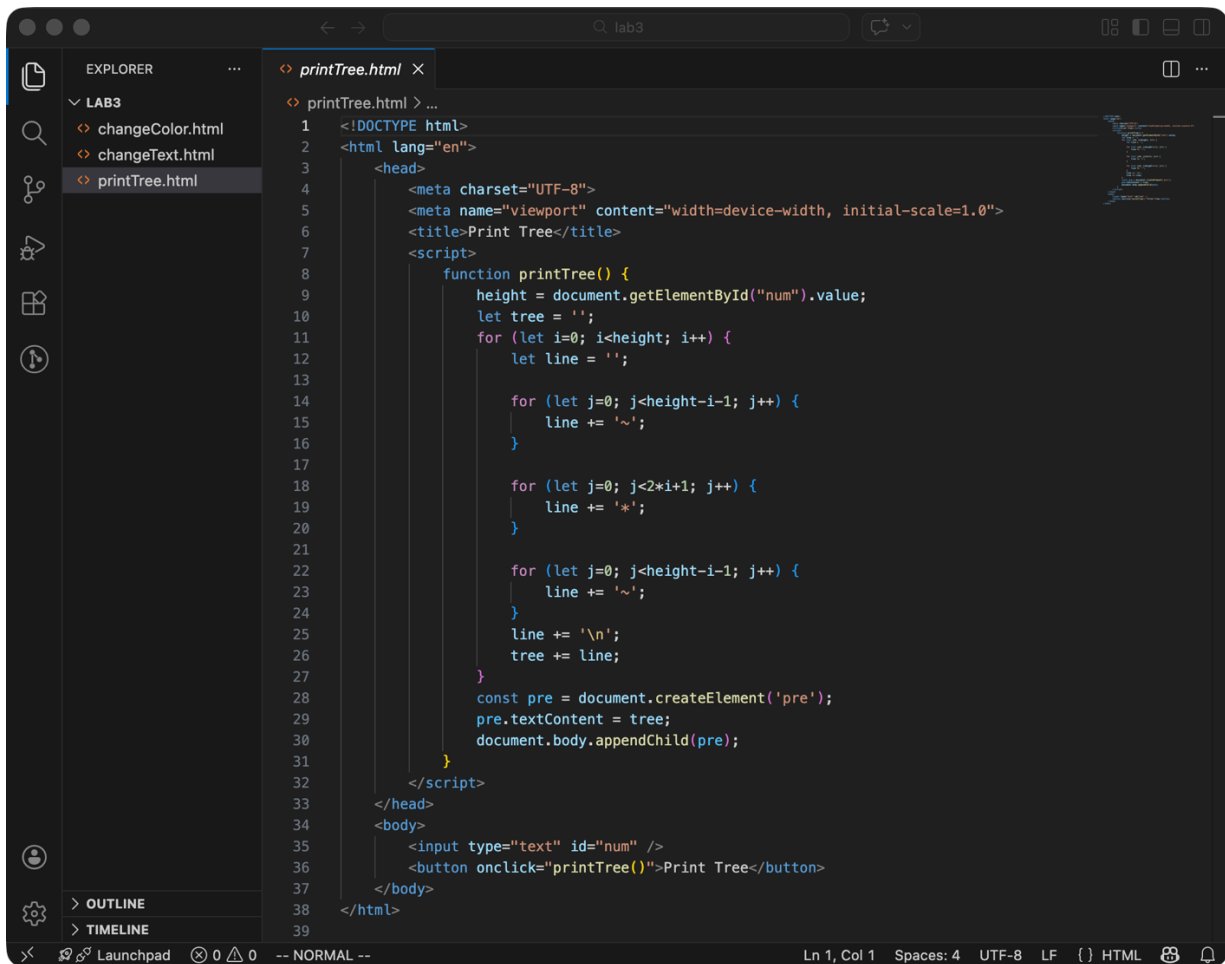
6. Access the webpage in your browser via URL: <http://localhost/can302/lab3/changeColor.html>



7. Click the button of "Change Color", it will change to:

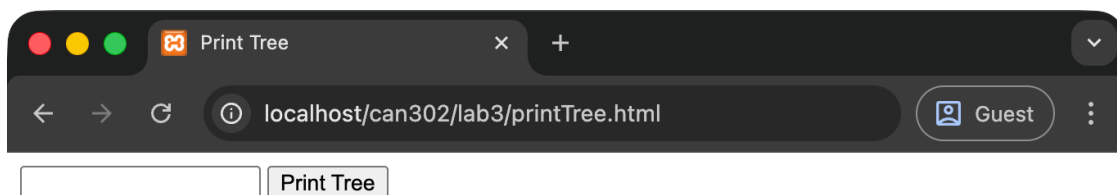


8. Create a new file "printTree.html" under "lab3". The code is to print a tree of given height:

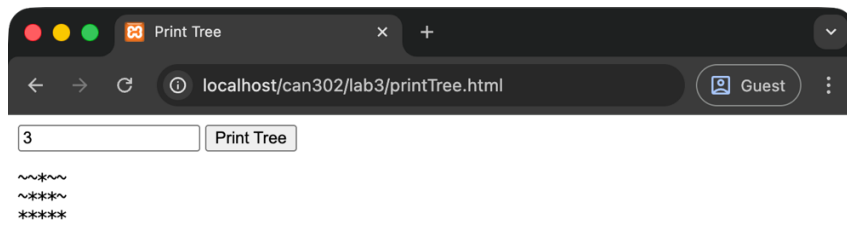


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Print Tree</title>
7     <script>
8       function printTree() {
9         height = document.getElementById("num").value;
10        let tree = '';
11        for (let i=0; i<height; i++) {
12          let line = '';
13
14          for (let j=0; j<height-i-1; j++) {
15            line += ' ';
16          }
17
18          for (let j=0; j<2*i+1; j++) {
19            line += '*';
20          }
21
22          for (let j=0; j<height-i-1; j++) {
23            line += ' ';
24          }
25          line += '\n';
26          tree += line;
27        }
28        const pre = document.createElement('pre');
29        pre.textContent = tree;
30        document.body.appendChild(pre);
31      }
32    </script>
33  </head>
34  <body>
35    <input type="text" id="num" />
36    <button onclick="printTree()">Print Tree</button>
37  </body>
38 </html>
39
```

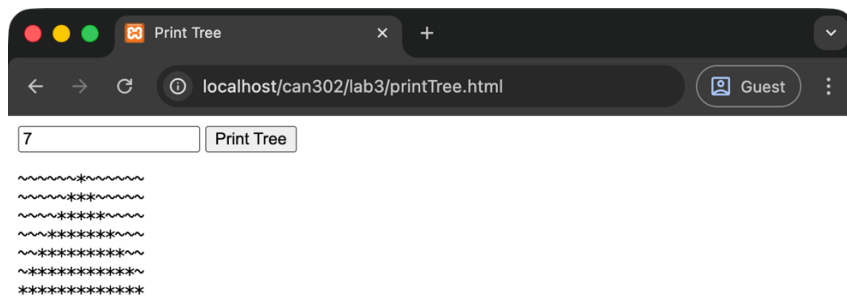
9. Access the webpage in your browser via URL: <http://localhost/can302/lab3/printTree.html>



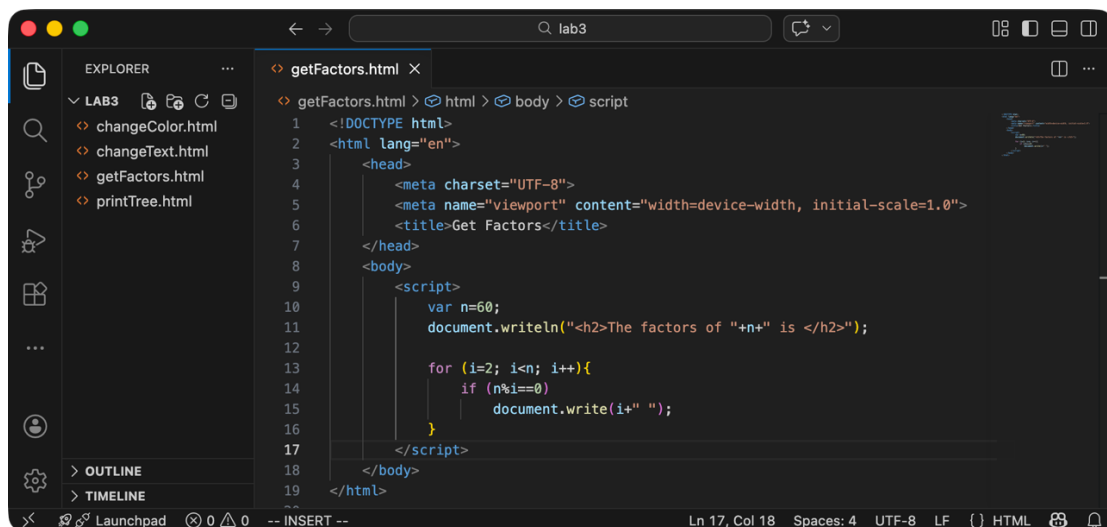
10. Input 3, and click the button of "Print Tree":



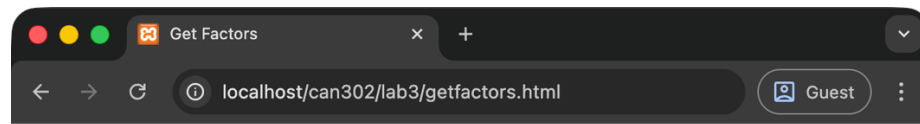
11. Reflash the webpage, input 7, and click the button of "Print Tree":



12. Create a new file "getFactors.html" under "lab3" as follows. The code is to print the factors of a given number n except 1 and itself. (You may modify the value of n to test your program)



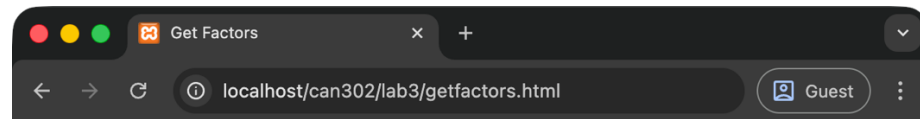
13. Access the webpage in your browser via URL: <http://localhost/can302/lab3/getFactors.html>



The factors of 60 is

2 3 4 5 6 10 12 15 20 30

14. Change the number of n to 256, and reflash the webpage:



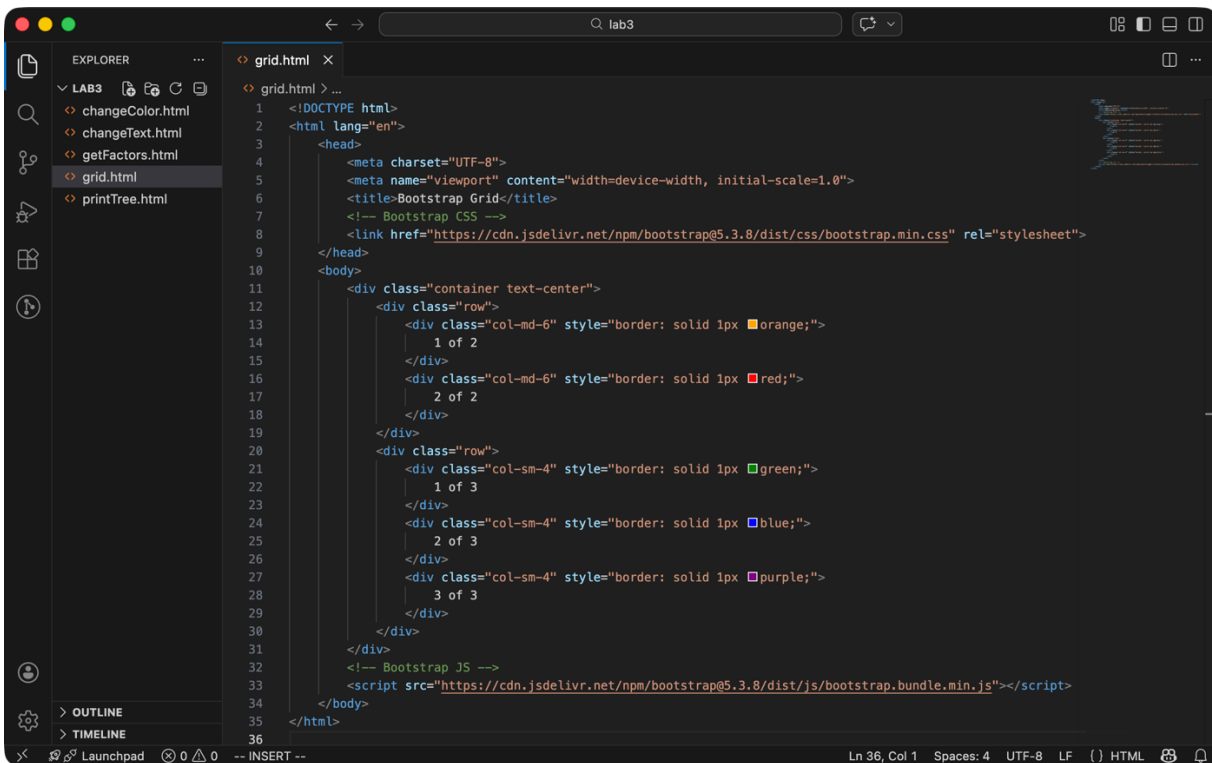
The factors of 256 is

2 4 8 16 32 64 128

B. Bootstrap Basics

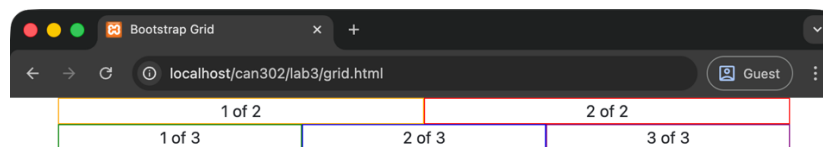
1. Create a new file "grid.html" under "lab3" as follows. You can divide the whole page into many rows and many columns in each row. To use bootstrap, you could download the JS/CSS files of bootstrap and include the corresponding JS/CSS files in your local folder (available at: <https://getbootstrap.com/docs/5.3/getting-started/download/>). You could also refer to the on-line one, as shown in the following example. The links to on-line bootstrap are:

- <https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css>
- <https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js>

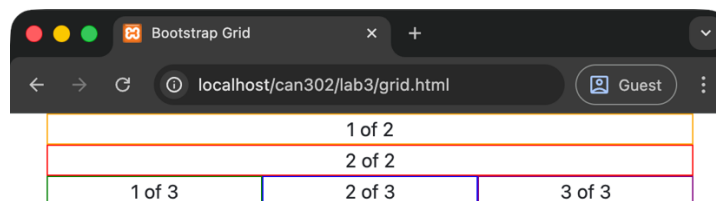


```
1 <!DOCTYPE html>
2 <html lang="en">
3
4   <head>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7     <title>Bootstrap Grid</title>
8     <!-- Bootstrap CSS -->
9     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet">
10  </head>
11  <body>
12    <div class="container text-center">
13      <div class="row">
14        <div class="col-md-6" style="border: solid 1px orange;">
15          1 of 2
16        </div>
17        <div class="col-md-6" style="border: solid 1px red;">
18          2 of 2
19        </div>
20      </div>
21      <div class="row">
22        <div class="col-sm-4" style="border: solid 1px green;">
23          1 of 3
24        </div>
25        <div class="col-sm-4" style="border: solid 1px blue;">
26          2 of 3
27        </div>
28        <div class="col-sm-4" style="border: solid 1px purple;">
29          3 of 3
30        </div>
31      </div>
32    </div>
33    <!-- Bootstrap JS -->
34    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"></script>
35  </body>
36 </html>
```

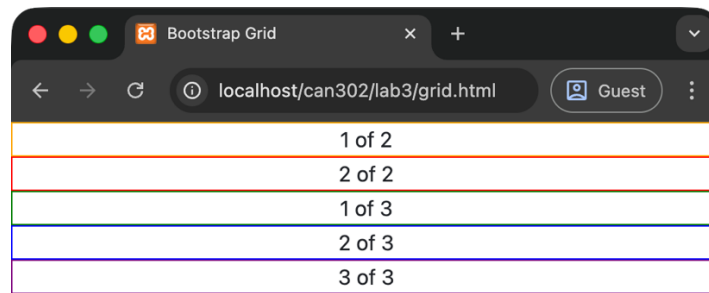
2. Access the webpage in your browser via URL: <http://localhost/can302/lab3/grid.html>



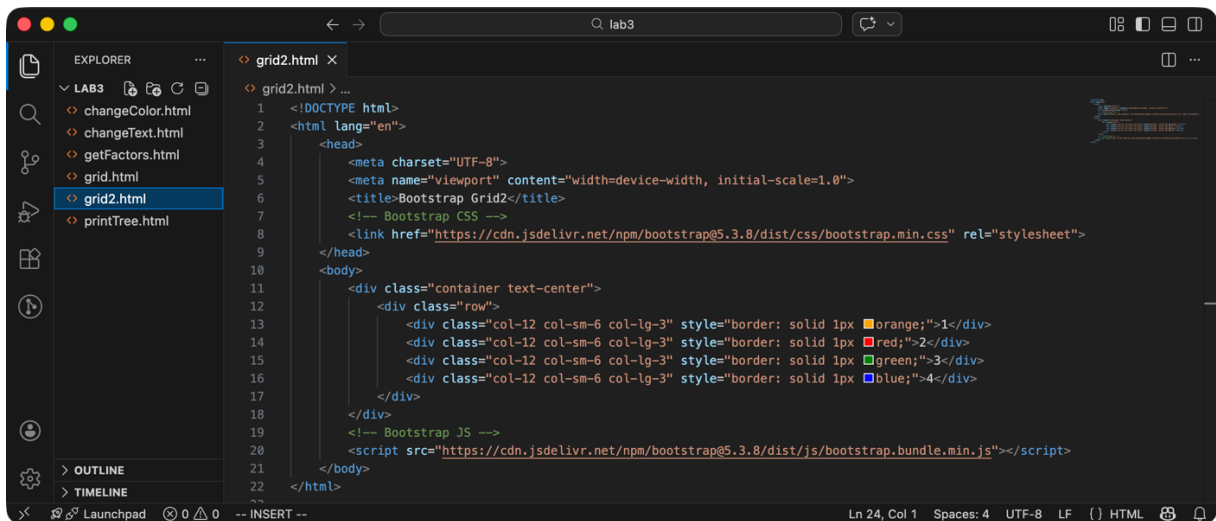
3. Try to reduce the width of your browser, then it looks as:



4. Try to further reduce the width of your browser, then it looks as:



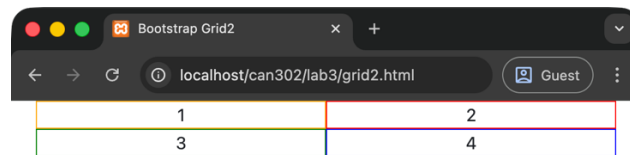
5. Create a new file "grid2.html" under "lab3" as follows:



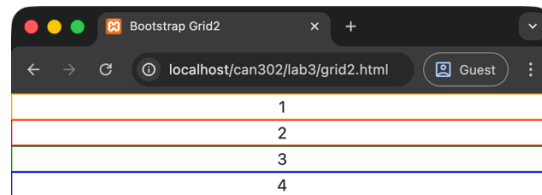
6. Access the webpage in your browser via URL: <http://localhost/can302/lab3/grid2.html>



7. Try to reduce the width of your browser, then it looks as:



8. Try to further reduce the width of your browser, then it looks as:

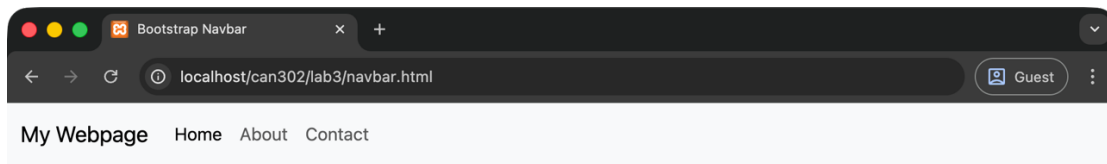


9. Create a new file "navbar.html" under "lab3" as follows:

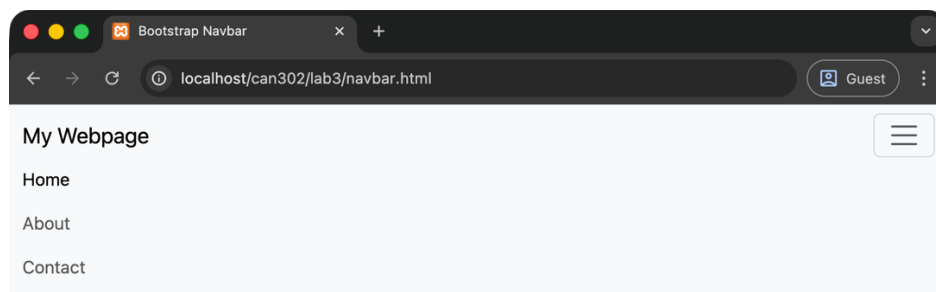
A screenshot of a code editor window showing the content of a file named "navbar.html". The file is located under a folder named "lab3". The code is as follows:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Bootstrap Navbar</title>
7     <!-- Bootstrap CSS -->
8     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet">
9   </head>
10  <body>
11    <nav class="navbar navbar-expand-lg navbar-light bg-light">
12      <div class="container-fluid">
13        <a class="navbar-brand" href="#">My Webpage</a>
14        <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
15          <span class="navbar-toggler-icon"></span>
16        </button>
17        <div class="collapse navbar-collapse" id="navbarNav">
18          <ul class="navbar-nav">
19            <li class="nav-item">
20              <a class="nav-link active" href="#">Home</a>
21            </li>
22            <li class="nav-item">
23              <a class="nav-link" href="#">About</a>
24            </li>
25            <li class="nav-item">
26              <a class="nav-link" href="#">Contact</a>
27            </li>
28          </ul>
29        </div>
30      </div>
31    </nav>
32    <!-- Bootstrap JS -->
33    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"></script>
34  </body>
35 </html>
```

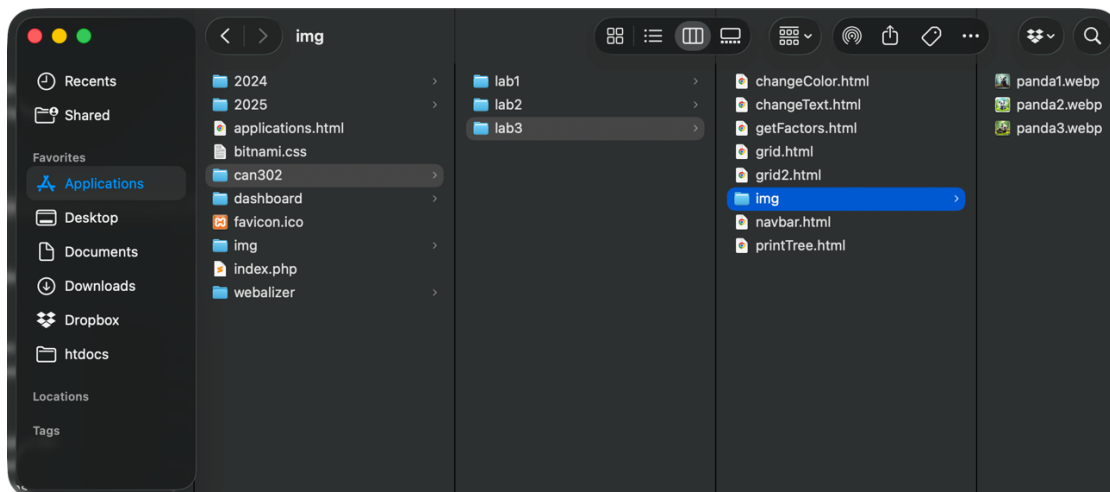
10. Access the webpage in your browser via URL: <http://localhost/can302/lab3/navbar.html>



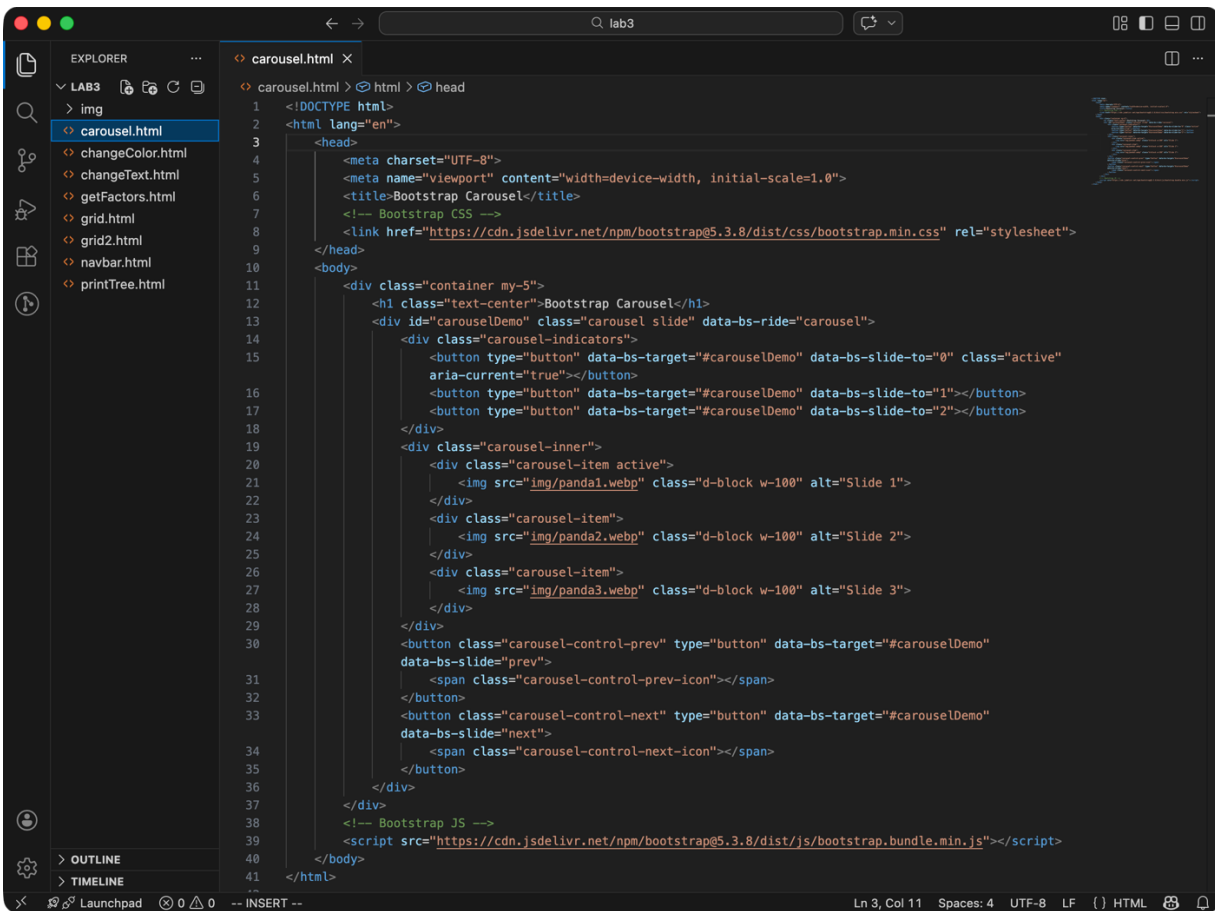
11. Try to reduce the width of your browser, then it looks as:



12. Create a subfolder "img" under "lab3", and put three pictures under it. For example:

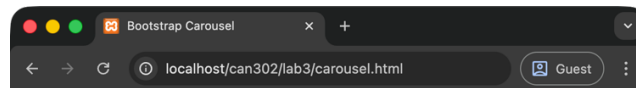


13. Create a new file "carousel.html" under "lab3" as follows. This example creates an image carousel with three slides and navigation buttons. You can click the buttons to view different slides. Remember to change the image names in your code (Lines 21, 24, and 27).

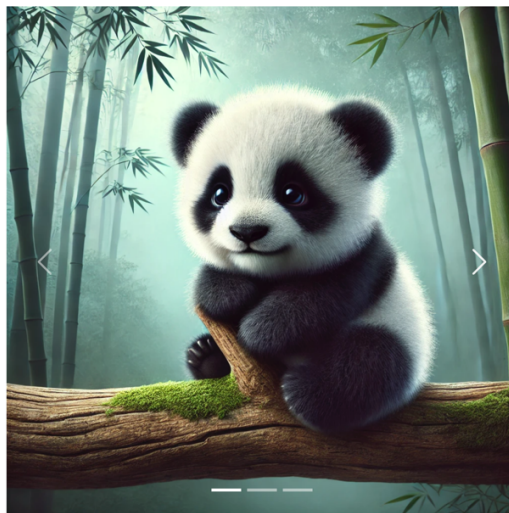


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Bootstrap Carousel</title>
7     <!-- Bootstrap CSS -->
8     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet">
9   </head>
10  <body>
11    <div class="container my-5">
12      <h1 class="text-center">Bootstrap Carousel</h1>
13      <div id="carouselDemo" class="carousel slide" data-bs-ride="carousel">
14        <div class="carousel-indicators">
15          <button type="button" data-bs-target="#carouselDemo" data-bs-slide-to="0" class="active"
16            aria-current="true"></button>
17          <button type="button" data-bs-target="#carouselDemo" data-bs-slide-to="1"></button>
18          <button type="button" data-bs-target="#carouselDemo" data-bs-slide-to="2"></button>
19        </div>
20        <div class="carousel-inner">
21          <div class="carousel-item active">
22            
23          </div>
24          <div class="carousel-item">
25            
26          </div>
27          <div class="carousel-item">
28            
29          </div>
30        </div>
31        <button class="carousel-control-prev" type="button" data-bs-target="#carouselDemo"
32          data-bs-slide="prev">
33          <span class="carousel-control-prev-icon"></span>
34        </button>
35        <button class="carousel-control-next" type="button" data-bs-target="#carouselDemo"
36          data-bs-slide="next">
37          <span class="carousel-control-next-icon"></span>
38        </button>
39      </div>
40    </div>
41    <!-- Bootstrap JS -->
42    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"></script>
43  </body>
44</html>
```

14. Access the webpage in your browser via URL: <http://localhost/can302/lab3/carousel.html>



Bootstrap Carousel



C. Hand-on Exercises

1. Practice DOM manipulation and event handling in JavaScript.

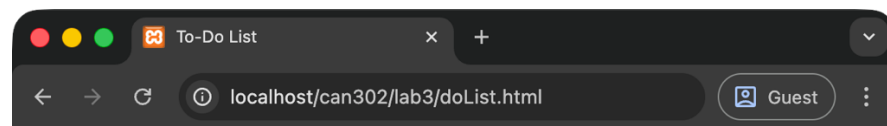
(a) Create an HTML page with:

- An input field to add a new task
- A button labeled "Add Task"
- An empty unordered list () to display the tasks

(b) Use JavaScript to:

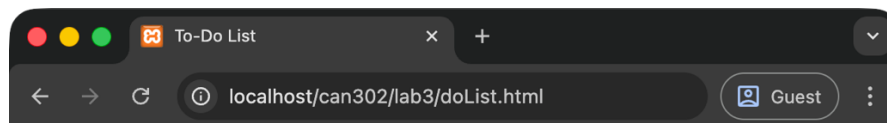
- Add a new task to the list when the "Add Task" button is clicked
- Display an alert if the input field is empty
- Add a "Delete" button next to each task to remove it from the list

(c) Examples:



To-Do List

Enter a new task Add Task



To-Do List

Enter a new task Add Task

- CAN302 Delete

2. Practice JavaScript functions and user input handling.

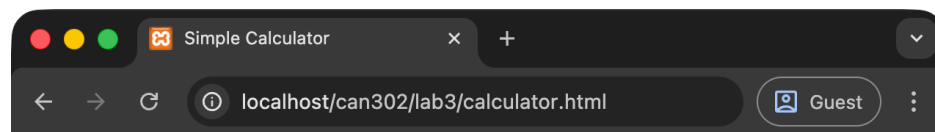
(a) Create an HTML page with:

- Two input fields for numbers
- Buttons for addition (+), subtraction (-), multiplication (*), and division (/)
- A paragraph to display the result

(b) Use JavaScript to:

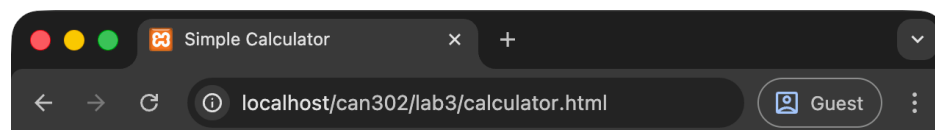
- Perform the selected operation when a button is clicked
- Display an alert when the denominator is zero
- Display the result in the paragraph

(c) Examples:



Simple Calculator

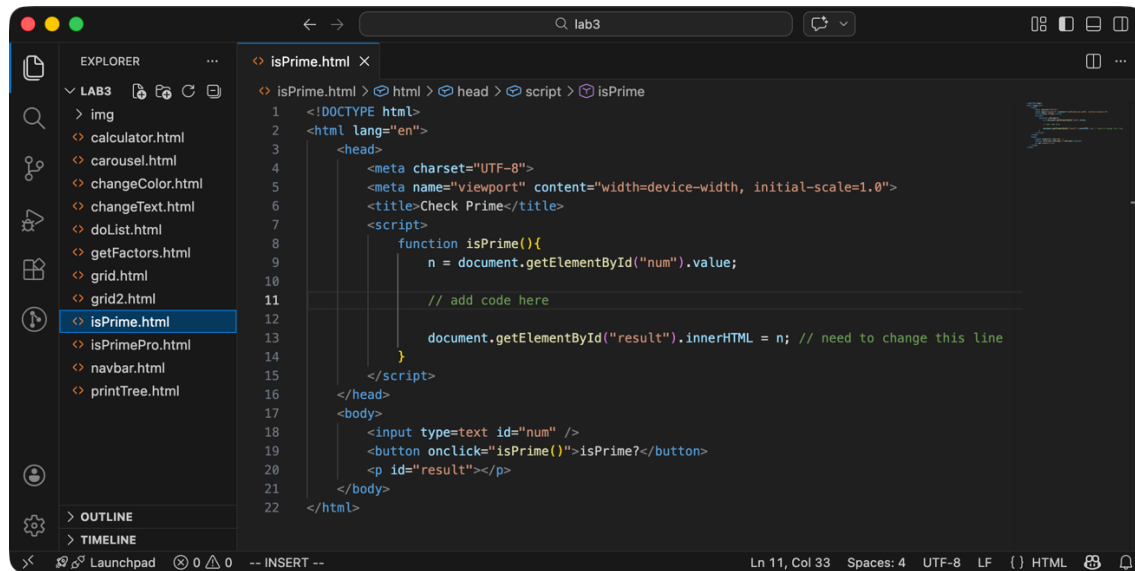
Result (+): 5



Simple Calculator

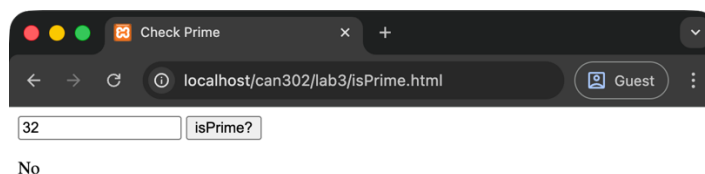
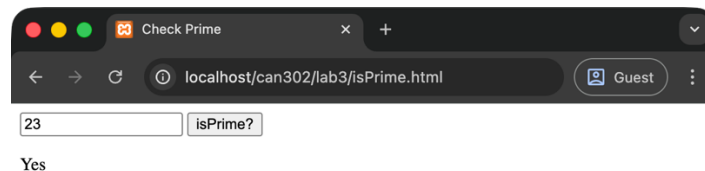
Error: Division by zero!

3. Practice JavaScript functions: Add JavaScript code to the following webpage such that it will determine the number in text box is a prime number or not. When user press the button, the browser will call the function "isPrime()" and modify the content of HTML element "result". Currently, the program just copies the user input to the result element.

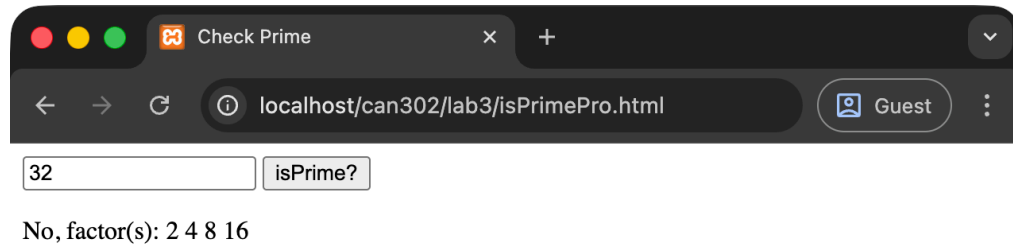


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Check Prime</title>
7     <script>
8       function isPrime(){
9         n = document.getElementById("num").value;
10
11         // add code here
12
13         document.getElementById("result").innerHTML = n; // need to change this line
14       }
15     </script>
16   </head>
17   <body>
18     <input type="text" id="num" />
19     <button onclick="isPrime()">isPrime?</button>
20     <p id="result"></p>
21   </body>
22 </html>
```

And examples:



4. Modify the program in Exercise 3 such that it would print the factors of the number if it's not a prime number. For example:



Summary

We have achieved the following things in this lab session:

- Write programs in JavaScript
- Handle HTML events with JavaScript
- Use Bootstrap to create responsive webpages